

Module 5: Modeling and Analysis of Bivariate Data

Essential Concepts

Background

The familiar slope-intercept form of a linear equation you probably learned about in algebra is where $y = mx + b$

$x \rightarrow$ represents the explanatory (independent) variable represented on the horizontal axis.

$y \rightarrow$ represents the response (dependent) variable represented on the vertical axis.

$m \rightarrow$ represents the constant rate of change (the slope)

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

$b \rightarrow$ represents the initial value in a given scenario (which can be zero or negative).

- the place on the graph where the line crosses the y -axis.

Example:

Kai weighed 173 pounds before Pam agreed to foster him. She modified his diet and began walking him three times daily. A reasonable goal was for Kai to lose about two pounds each week. For Kai's situation, identify the following:

(a) Explanatory variable?

- The number of weeks
- Kai's weight

(c) Slope: _____ pounds per week

(b) Response variable?

- The number of weeks
- Kai's weight

(d) y -intercept: _____ pounds

Scatterplots

Scatterplots: used to illustrate the relationship between two **quantitative** variables.

Such data from two quantitative variables (usually two related data) are called **bivariate** data.

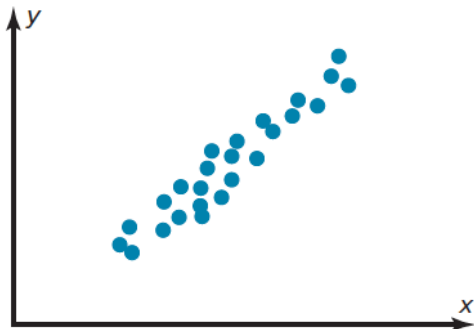
When investigating relationships between two quantitative variables, scatterplots are a simple way to visually represent the **spread**, **direction**, **strength of relationship**, and **potential outliers** of the data.

To create a scatterplot, we use an **ordered pair** to represent the data.

The x -coordinate is the **explanatory** variable, and the y -coordinate is the **response** variable.

Ways to **describe** data:

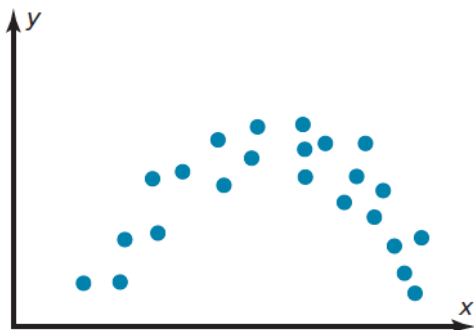
- direction (positive or negative)
- form (linear or non-linear)
- strength (strong or weak)



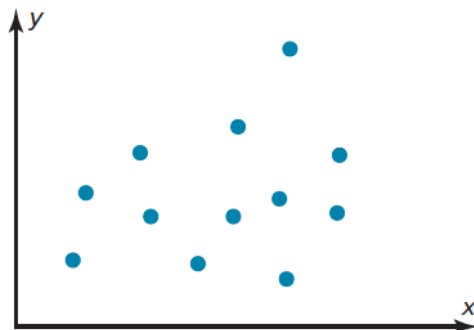
(a) Positive linear relationship



(b) Negative linear relationship

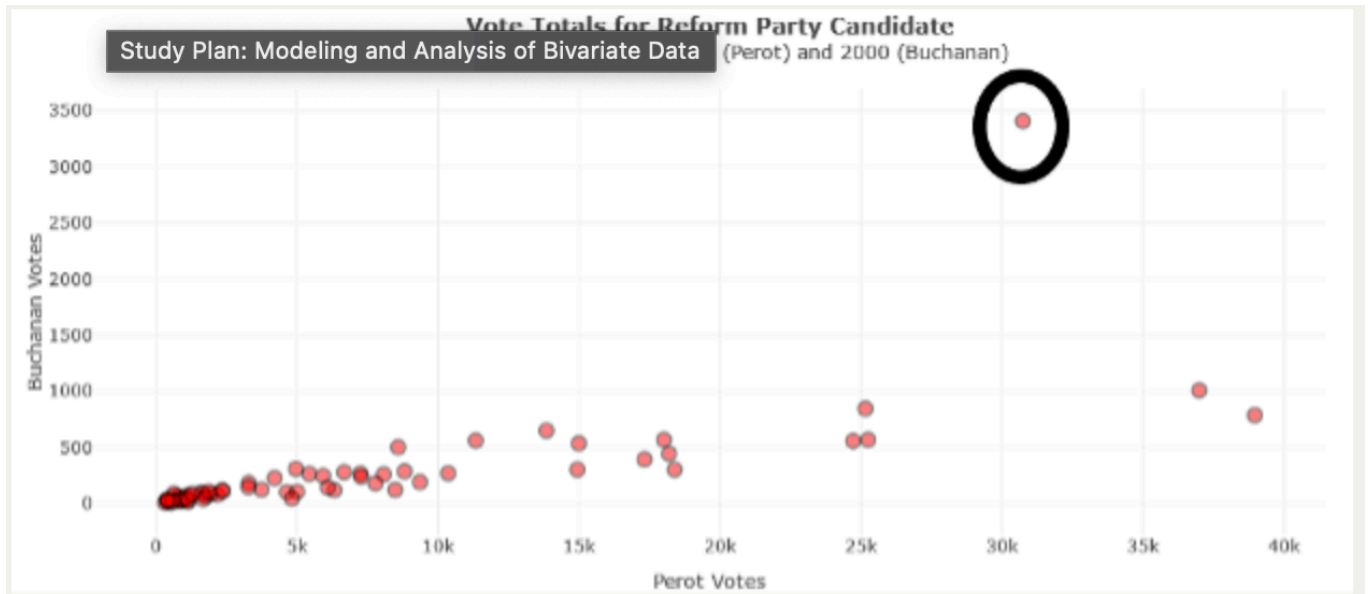


(c) Curvilinear relationship

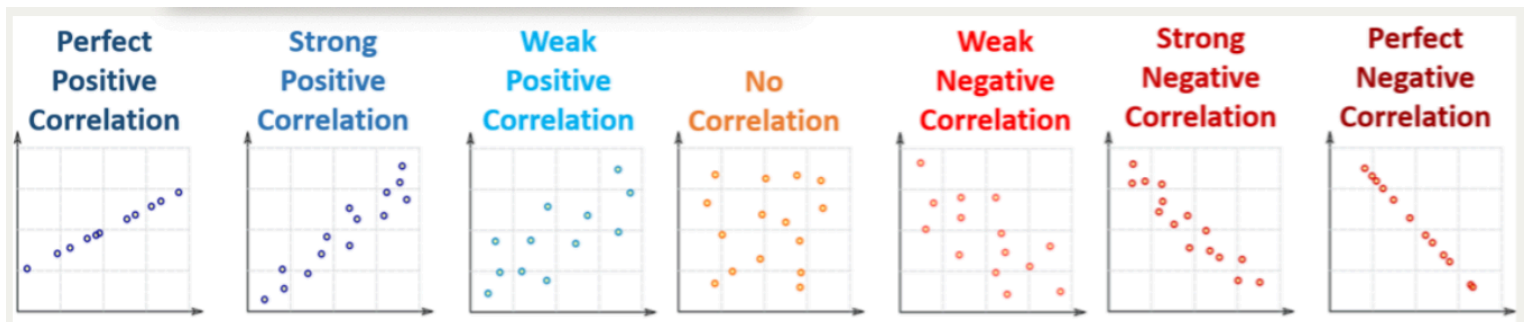


(d) No relationship

Outliers appear as departures from the general trend.



To describe **trends**:



Correlation Coefficient:

The **correlation coefficient**, r , is a numeric measure that measures the **strength** and **direction** of a **linear** relationship between two quantitative variables.

Formula:
$$r = \frac{\frac{x - \bar{x}}{s_x} \cdot \frac{y - \bar{y}}{s_y}}{n - 1}$$
, for our purposes mainly concerned with the interpretation of r .

Correlation Coefficient, r	General Interpretation
Study Plan: Modeling and Analysis of Bivariate Data	
−1 to −0.7	Strong negative linear relationship
−0.7 to −0.3	Moderate negative linear relationship
−0.3 to −0.1	Weak negative linear relationship
−0.1 to 0.1	Negligible or no linear relationship
0.1 to 0.3	Weak positive linear relationship
0.3 to 0.7	Moderate positive linear relationship
0.7 to 1	Strong positive linear relationship

Properties of r :

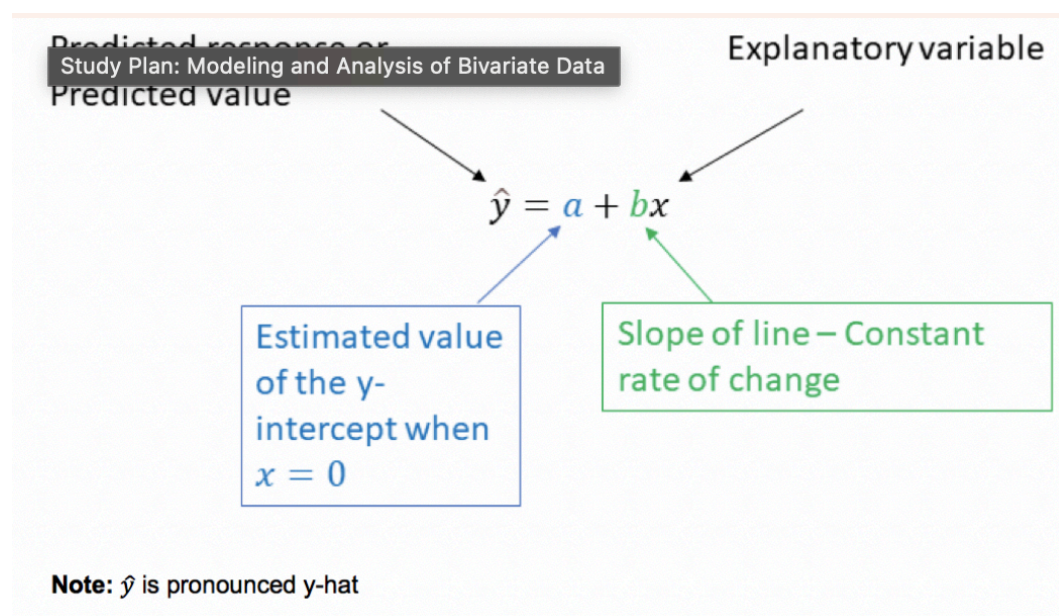
- The correlation does not change when the units of measurement of either one of the variables change.
- The correlation measures only the strength of a linear relationship between two variables.
 - Always make a scatterplot of the data before interpreting the meaning.
- Association does not imply causation

Line of Best Fit

Line of best fit: the best line that describes the data points.

- For real data with natural deviations, the line cannot go through all of the points.
- Goal is to **minimize error**.
 - Where it gets its name: **Least Squares regression Line (LSRL)**

Equation of LSRL:



We will be interpreting the **estimated or predicted slopes** and **estimated or predicted intercepts** generated from given data sets. If the data changes, these values will change.

estimated or predicted slopes: $a = \bar{y} - b\bar{x}$

estimated or predicted intercepts: $b = r \cdot \frac{s_y}{s_x}$

Template for interpreting slope:

For every one [unit] increase in **[explanatory variable units]**, we predict an average **increase/decrease** of ____ **[response variable units]** in **[response variable]**.

Example: For every increase of one chirp per second, the estimated temperature increases on average by 3.29 degrees Fahrenheit.

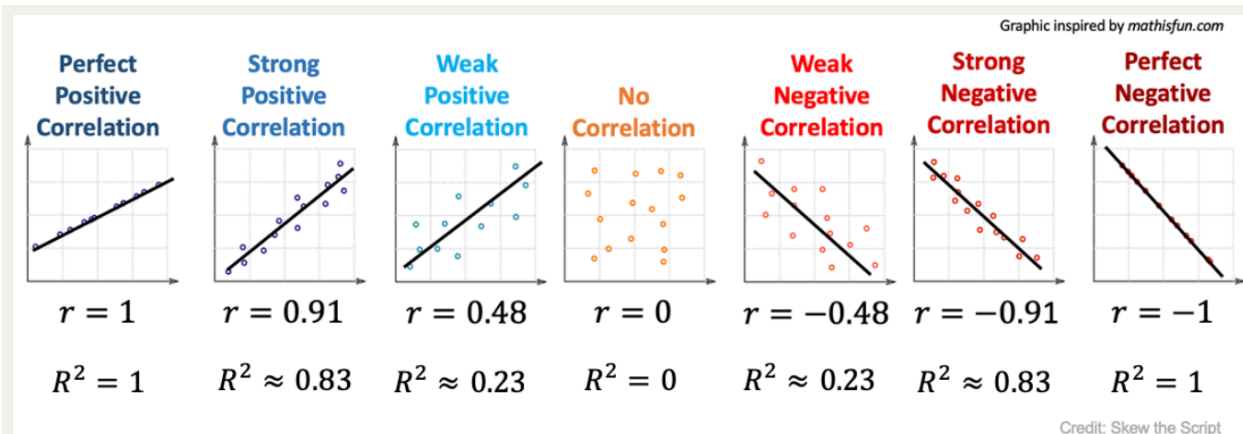
Extrapolation:

Prediction for values of the explanatory variable that fall outside the range of the data is called **extrapolation**.

These predictions are **unreliable** because we do not know if the pattern observed in the data continues outside the range of the data. **Avoid making predictions outside the range of the data!**

Coefficient of Determination

Coefficient of determination: denoted or R^2 (pronounced “r- squared”), is the proportion of the variation in the response variable that can be explained by its linear relationship with the explanatory variable.



Properties of R^2 :

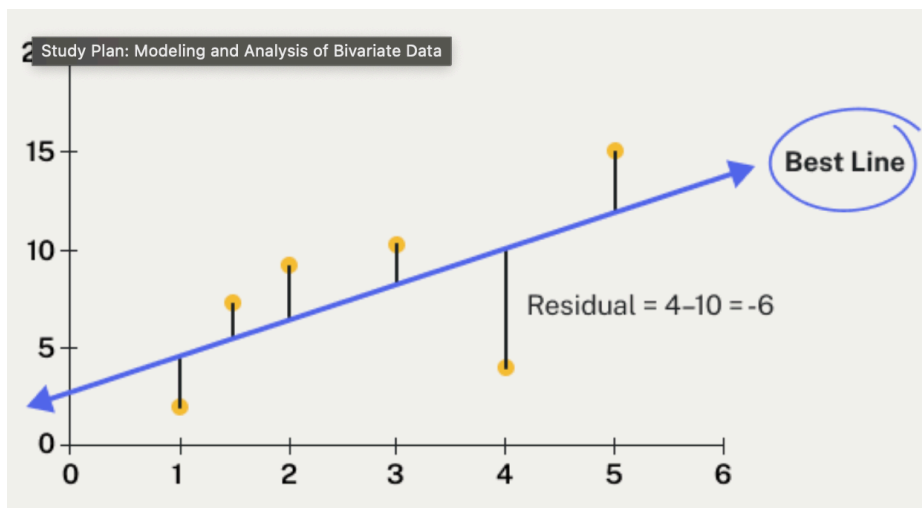
- R^2 : the coefficient of determination is equal to the square of the correlation coefficient. Because of this, R^2 is more sensitive to differences in the strength of the linear relationship between the two variables than r is.
- Will always be a positive number between 0 % and 100 % .
- Looking to see how well the models explains the variation of the response variable

Residuals

Residual for a data point is the **difference** between the **observed** value of the response variable and the linear model's **prediction**.

$$residual = y - \hat{y}$$

- vertical error associated with each data point
- represents how far off a prediction calculated from the line is compared to the actual, observed value
 - the **larger** the line, the **greater** the error associated with that particular observation.
 - For data points that are above the line of best fit, the residuals are **positive**,
 - For data points that are below the line, the residuals are **negative**.



We might worry about the appropriateness of the model if we notice the following:

- The trend in the scatterplot is **non-linear**, indicating that the relationship between the explanatory variable and the response variable is not modeled very well by a line.
- The residuals tend to have a **pattern**. The observations are above and below the line systematically.
- The observed values are further and further **away from the line of best fit** for a portion of the data. That is, the errors are not consistent for all values of the explanatory variable.

Residual standard error, s_e : measure of the variability in the residuals

A large residual standard error indicates there is a lot of spread in the scatter of the data.